



ENGINEERING MATHEMATICS PROJECT

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From Noise to Clarity: A Study of Filtering Techniques in Biomedical Imaging

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1 Aspects of an Image

In this part we want to investigate through different properties of an image. In image processing, analyzing different aspects helps us understand and manipulate visual information. Among these aspects we can mention Magnitude, Phase and Frequency domain. To gain a better understanding of this issue, experiments will be conducted.

1.1 Questions

Using [pic1.jpg](#) answer questions below:

1. Intuitively, What do low and high frequencies of an arbitrary image contain? Which one is more important? Do you think your answer depends on the image?

The high frequencies of an arbitrary image contain the rapid intensity changes in the image like the edges; however, the lower frequencies define the more smooth features and textures with the lower intensity change like the backgrounds.

2. Explain about low pass filter(LPF) and high pass filter(HPF) . In what applications LPF and HPF are used? Write two functions that apply LPF and HPF to an arbitrary image.

The high pass filter is a function which pass the frequencies with higher magnitude and emitted the low frequencies, meanwhile the low pass filter act in reverse. both filters are used in various fields such as image processing , audio processing and electronics. 3. Apply LPF and HPF to [pic1.jpg](#) using functions written in the previous part. Explain the results.



Figure 1: Low pass and High pass filters

Using [pic2.jpg](#) and [pic3.jpg](#) answer questions below:

- 1) Plot the magnitude and phase of both images. Then, plot an image with magnitude of [pic2.jpg](#) and phase of [pic3.jpg](#). Plot the other alternate.

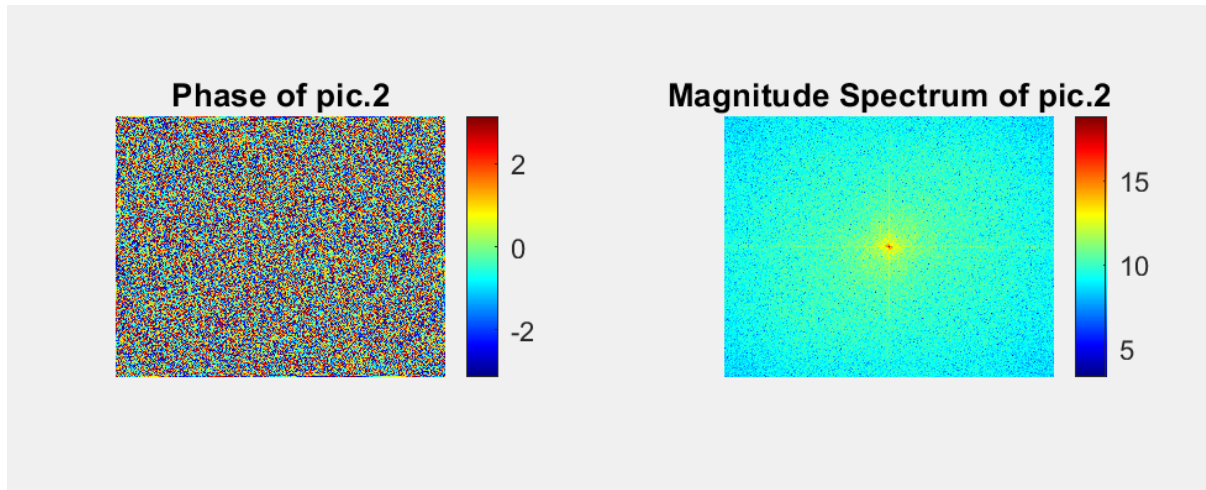


Figure 2: Pic2 features

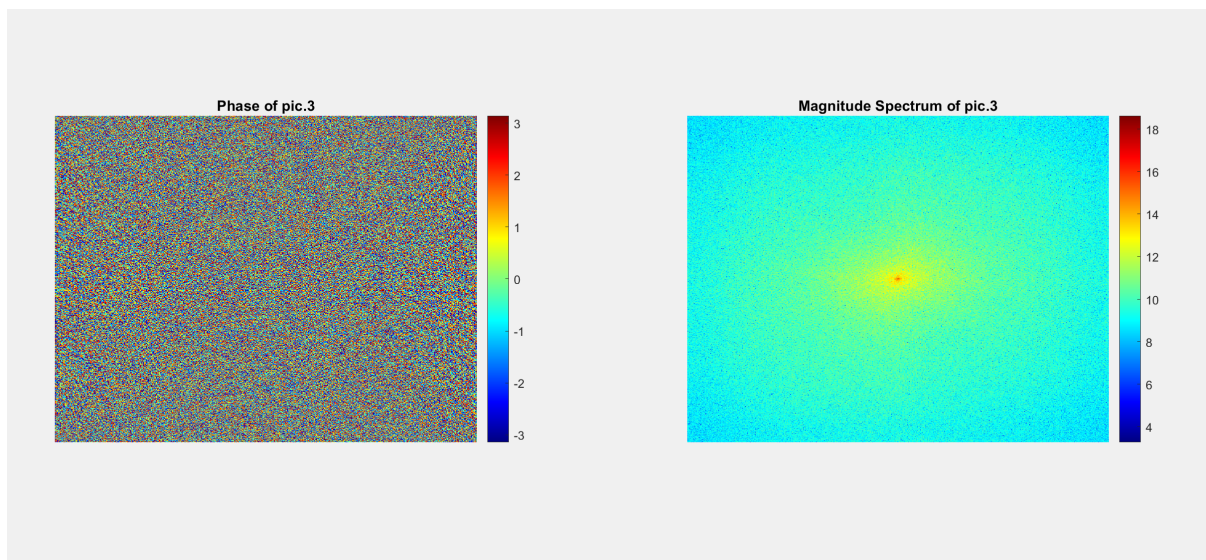


Figure 3: pic3 features

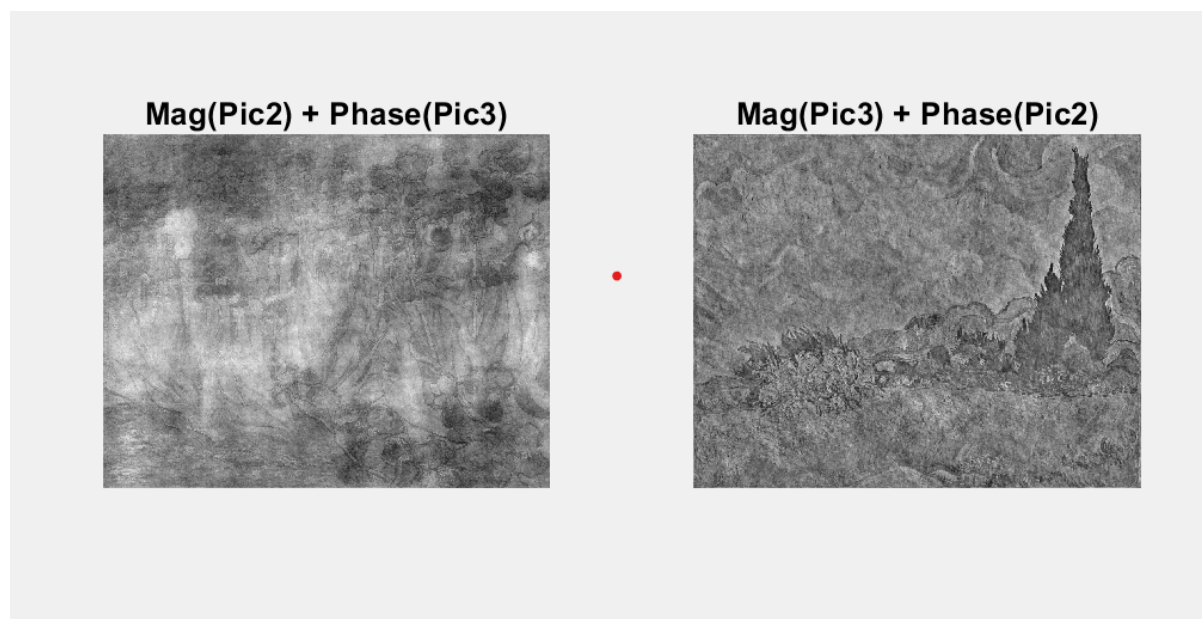


Figure 4: pic3 features

2) Using the results above, Explain Why is the phase information more critical for reconstructing recognizable features of an image compared to the magnitude information?

Phase spectrum is for recognizing objects because it defines the spatial structure and edges of the image. Magnitude spectrum mainly affects contrast and brightness but does not determine the structure. This is why, when swapping magnitude information between two images but keeping the phase of one, the output still resembles the original image whose phase was retained.

2 Noise

In this part we explore more about Noise. Noise in digital images and signals is an unwanted disturbance that affects quality and can originate from various sources. It can appear in different forms, such as salt-and-pepper noise, Gaussian noise and 50 HZ noise. Each type of noise requires specific filtering techniques for reduction, and understanding the nature of noise is crucial to effectively restoring signal or image quality. The following questions will help you better track the rest of the project.

2.1 Questions

1. What are the primary sources of salt-and-pepper noise in digital images, and what filtering techniques are most effective in mitigating this type of noise?

Data corruption during transmission over wireless or wired networks. Bit errors due to weak signal, interference, or compression artifacts. Dust or environmental factors affecting the image capture. Faulty memory locations in image storage or RAM corruption. Median Filter, Adaptive Median Filter, Mean Filter (Averaging Filter) are useful filters for this noise.

2. How does electrical interference introduce 50Hz noise into signal recordings, and what methods can be employed to eliminate or reduce this interference?

Electrical interference at 50Hz is caused by the power line frequency, which is caused by from the AC mains supply. Nearby power cables create an alternating magnetic field, inducing unwanted voltage in signal wires. Electrical noise from AC power lines can couple to signal lines through parasitic capacitance. ways to eliminate the noise; Low-Pass Filter: Can remove high-frequency noise but may not be ideal if useful signals exist near 50Hz. Fourier Transform (FFT-based Filtering): Identifies the 50Hz peak and removes it.

3. Identify the common sources of Gaussian noise in imaging systems and discuss the most suitable algorithms for its removal without significantly degrading image quality.

The common sources of Gaussian noise are: Shot Noise: Caused by the random arrival of photons at the sensor, leading to intensity fluctuations. Thermal Noise (Dark Current Noise): Due to random electron movement in the sensor, even in complete darkness. ways to eliminate the noise; Gaussian Smoothing (Basic) and bilateral filter are ways to eliminate the noise.

4. Explore SNR and PSNR. What roles do Signal-to-Noise Ratio (SNR) and Peak Signal-to-Noise Ratio (PSNR) play in evaluating the effectiveness of noise removal techniques in digital images? Compare and contrast the effectiveness of SNR and PSNR in different applications such as medical imaging, video compression, and wireless communications.

Signal-to-Noise Ratio (SNR) is a measure of how much a signal is corrupted by noise. It is defined as the ratio of the signal power to the noise power.

Peak Signal-to-Noise Ratio (PSNR) is a more specific measure of noise degradation in images and videos. It compares the peak possible signal value to the mean squared error (MSE)

The usage of the SNR is in general signal qualities including audio and the communications. SNR is less sensitive to small pixel level variation than PSNR. PSNR is used in image and video processing and compression.

2.2 Noise Modeling

2.2.1 Function Definition

Write a function named `noise` that takes three parameters a , L , and `type`, and works as follows:

- If the `type` input is the string "uniform", the function returns a vector of length L where each component is a random variable from the distribution $U(-a, a)$.
- If the `type` input is the string "normal", the function returns a vector of length L where each component is a random variable from the distribution $N(0, a^2)$.

2.2.2 Statistical Analysis of Noise Vectors

Before adding noise to the original signal, we observe some statistical properties of the noise and compare them with theoretical expectations.

(a) Generate and Plot Noise Vectors

Generate two noise vectors N_1 and N_2 with $a = 0.2$, using uniform and normal distributions respectively, with $L = 10000$. Plot the histogram of each vector (using MATLAB's `histogram` function). Do the resulting shapes confirm the expected probability distributions?

The figure 4 depicts the difference of the two types of generated noises

(b) Mean of Noise Vectors

For the noise vector N_1 , define:

$$N_1 = \frac{1}{L} \sum_{i=1}^L N_1[i]. \quad (5)$$

Compute N_1 for $L \in \{1, 2, \dots, 1000\}$ and plot it as a function of L . Repeat the same for N_2 . What is the limit of N_1 and N_2 as $L \rightarrow \infty$? Which theorem or statement in probability theory precisely explains this result?

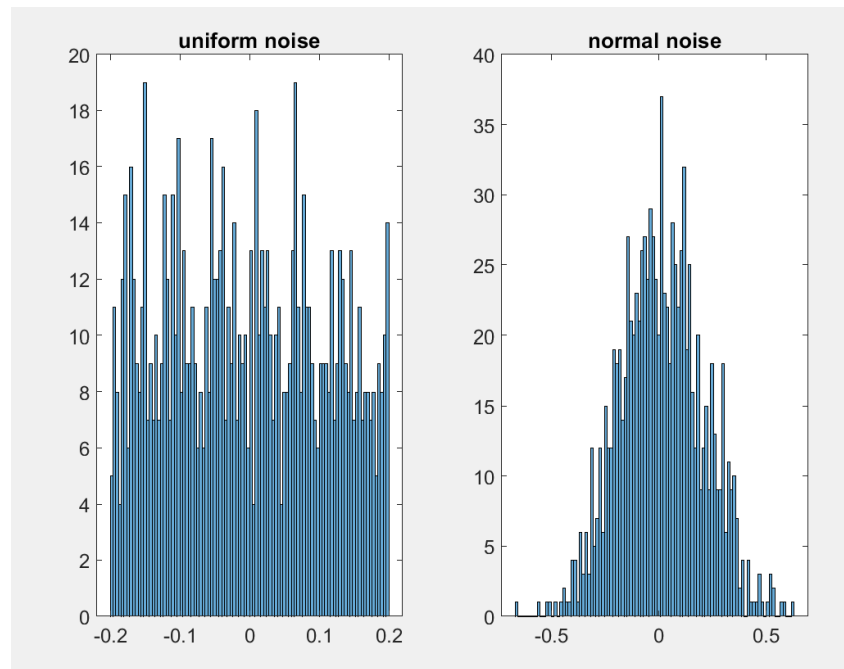


Figure 5: The difference between the uniform and random noise generator

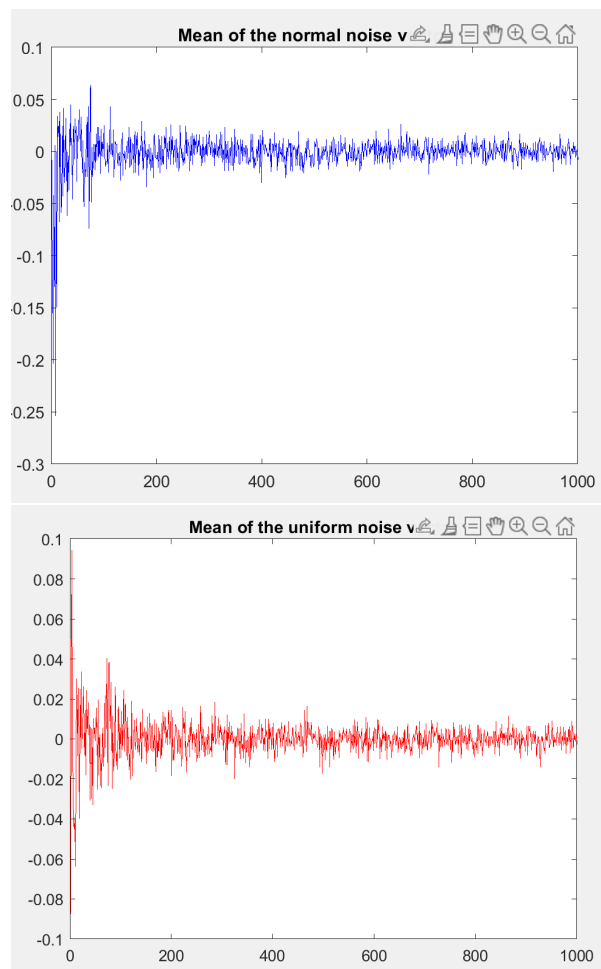


Figure 6

As L increases, the values of N_1 and N_2 will converge to their expected values (which are 0 for Gaussian noise with zero mean). The Law of Large Numbers guarantees this convergence. Specifically:

If $N_1[i] \sim N(0, 1)$ and $N_2[i] \sim N(0, 1)$, we expect:

$$\lim_{L \rightarrow \infty} N_1(L) = 0 \quad \text{and} \quad \lim_{L \rightarrow \infty} N_2(L) = 0.$$

(c) Energy of Noise Vectors

Consider the following definition of the energy of a noise vector N :

$$E_N = \frac{1}{L} \sum_{i=1}^L (N[i])^2. \quad (6)$$

For each of the two noise vectors N_1 and N_2 , compute the energy as a function of L for $L \in \{1, 2, \dots, 1000\}$, and plot it. What is the limit of E_{N_1} and E_{N_2} as $L \rightarrow \infty$? Verify your result through simulation. The amount of energy for both noise is 0;

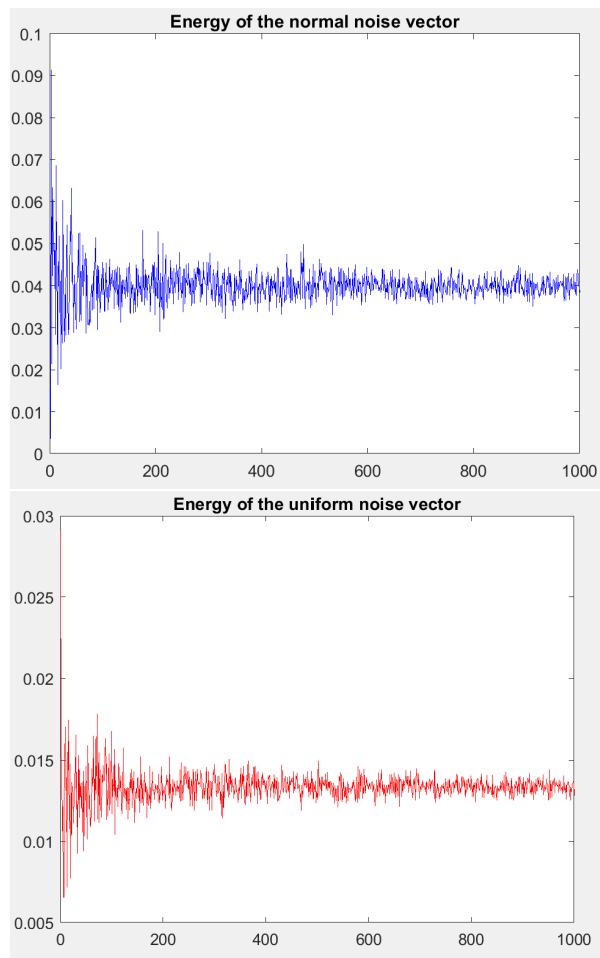


Figure 7

(d) Cross-Correlation of Noise Vectors

For the two noise vectors N_1 and N_2 , define:

$$c_{12} = \frac{1}{L} \sum_{i=1}^L N_1[i]N_2[i]. \quad (7)$$

Plot c_{12} as a function of L for $L \in \{1, 2, \dots, 1000\}$. Also, compute the limit of c_{12} as $L \rightarrow \infty$ and verify it through simulation. Explain what information c_{12} provides about the two vectors. Justify how c_{12} serves as a measure of similarity.

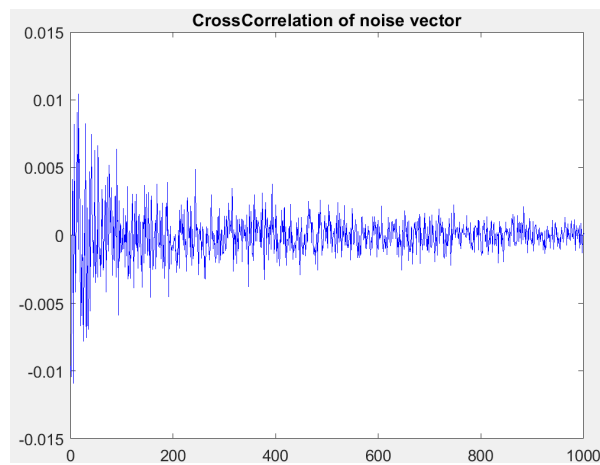


Figure 8

Cross-correlation is a measure of similarity because it quantifies how much the two vectors move together on average. If c_{12} is close to zero, it indicates that the vectors are largely uncorrelated. A positive or negative value for c_{12} would indicate some degree of similarity or anti-similarity, respectively.

Conclusion:

2.2.3 Sinusoidal Signal and Noise Addition

In this question, the original signal used is a simple sinusoidal signal defined as:

$$x(t) = \sin(2\pi f_0 t), \quad f_0 = 1 \text{ Hz}.$$

The time range of the signal is considered as $[0, 2]$. Generate the signal $x(t)$ with an appropriate sampling frequency such that the plotted graph in MATLAB closely resembles a continuous sinusoidal curve. Plot $x(t)$ and specify the sampling frequency used in the report.

Next, add the noise vectors N_1 and N_2 to the generated signal to produce two noisy signals x_{n1} and x_{n2} :

$$x_{n1}(t) = x(t) + N_1(t), \quad x_{n2}(t) = x(t) + N_2(t). \quad (8)$$

Plot both noisy signals in the report.

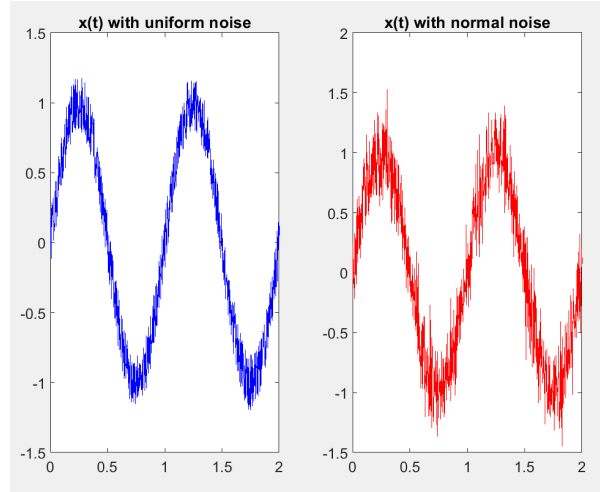


Figure 9

2.2.4 Energy Function Definition

Write a function to compute the energy of a vector x , as defined by equation (6):

$$E = \text{energy}(x),$$

where the function takes a vector as input and outputs its energy.

2.2.5 Energy Computation of Signals

Using the `energy` function written in the previous section, compute the energy of the following signals:

- The original signal x .
- The noise vectors N_1 and N_2 .
- The noisy signals x_{n1} and x_{n2} .

In total, compute the energies of five signals. Report the computed energy values. Verify

whether the relation $E_{x_n} = E_x + E_N$ approximately holds. For any two signals x and y , determine if the statement $E_{x+y} = E_x + E_y$ is true or false. Prove or disprove the statement and validate it for this specific example. As we can see in Figure 9

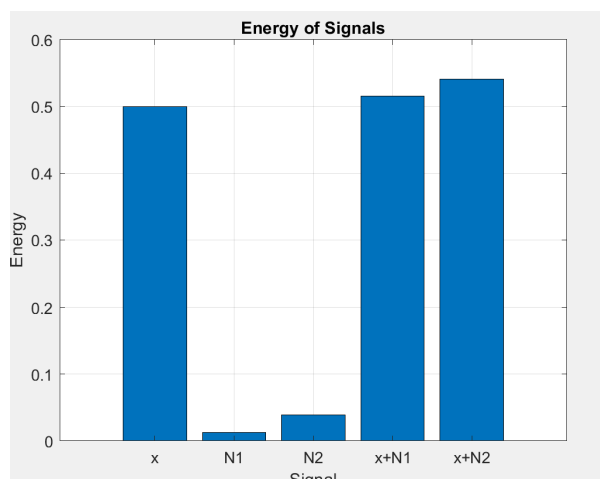


Figure 10

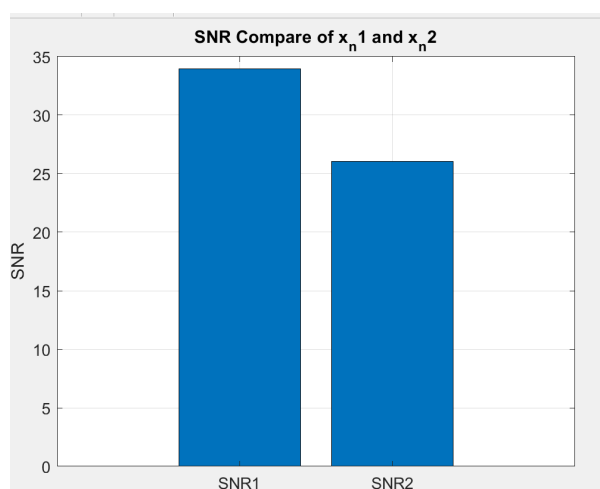
2.2.6 Signal-to-Noise Ratio (SNR)

The ratio of the energy (power) of the original signal to the energy (or power) of the noise is called the Signal-to-Noise Ratio (SNR):

$$\text{SNR} = \frac{E_x}{E_N}.$$

Compute and report the SNR for the signals x_{n1} and x_{n2} generated in Section 3.2.3.

Subsequently, increase the value of a in the noise definitions until the smallest value of a is found such that there is no noticeable visual similarity between x and x_n . Report the corresponding value of a and the SNR for each type of noise separately.



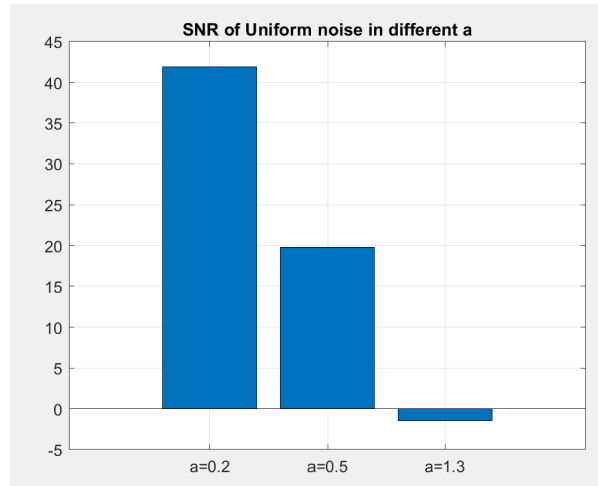


Figure 11: The effect of a value in SNR

3 Magnetic Resonance Imaging (MRI): A Comprehensive Overview

3.1 Questions

Explore the main sources of noise in this method and suggest some ways to deal with this problem.

- **Thermal Noise:** the shape of the noises are mostly like salt and pepper and grainy. This noise is the result of random electronic fluctuations in the MRI. Median filter and Gaussian filter are useful for detecting noises.
- **Motion Artifacts:** The noises are Blurry streaks, ghosting, or wavy distortions. and this noise is caused by patient movement while MRI scans. One of the ways for filtering this noise is Fourier Transform-Based Filter which identifies and removes motion-induced frequency components.
- **RF interference:** the noised image includes Bright horizontal or vertical stripes, wave-like patterns. External radiofrequency (RF) signals interfering with MRI signals is the reason of this type of the noise. Some ways for filtering this noise: Notch Filtering in Frequency Domain: Removes specific RF noise frequencies using Fourier Transform techniques. Adaptive Noise Cancellation: Algorithms that separate the noise signal from the MRI data and subtract it.

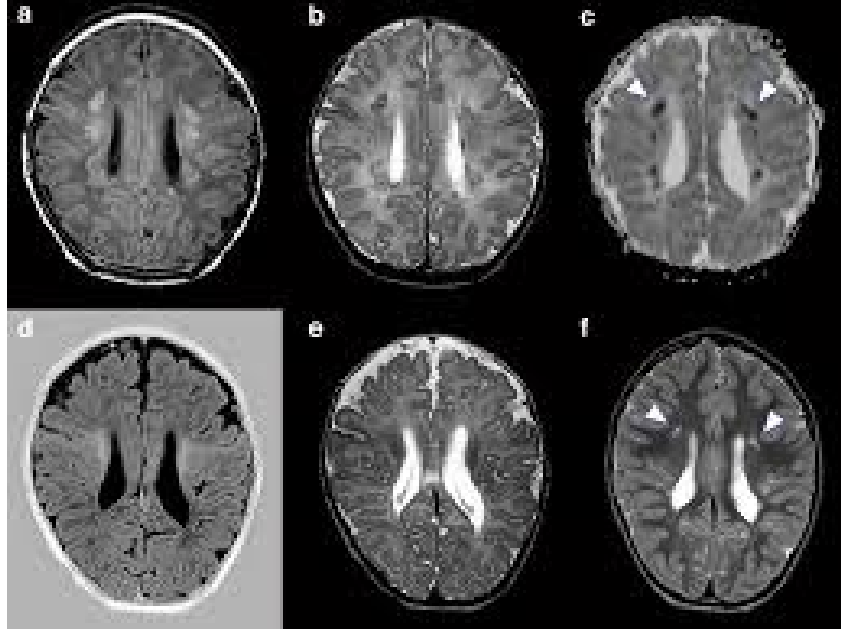


Figure 12: Brain MRI

4 Filters

In this part we learn about different filters, how they work and what's their impact on image. Then using these filters, you denoise two given noisy images and measure how effective each filter is.

Notice: You are not allowed to use built-in functions for this part.

4.1 Mean Filter

Investigate the Mean filter and implement a function to apply it to an input image. Explain about the effect of kernel size.

MEAN FILTER is a linear spatial filter used to reduce and enhance the noise and clarity of the input image. It is more beneficial to reduce the intensity variation among the neighboring pixels. That is, the pixels of each image are replaced with the mean value of the neighboring pixels.

$$\text{filtered_image}(i, j) = \frac{1}{(2k+1)^2} \sum_{l=-k}^k \sum_{e=-k}^k \text{image}(i+l, j+e)$$

As we can see in figure 4, this filter won't depict the edges and intensities completely by increasing the kernel size. But in figure 5 we can see how the mean filter works properly for the grainy noises. Now we know that we can use this filter for the thermal noise type in MRI pictures.

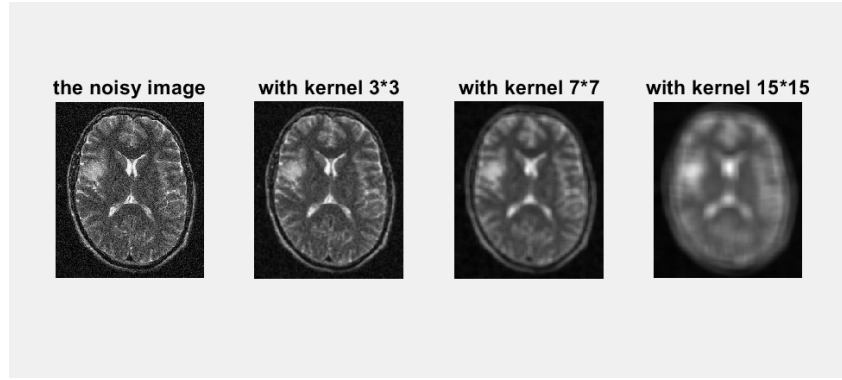


Figure 13: the effect of kernel size in noisy image

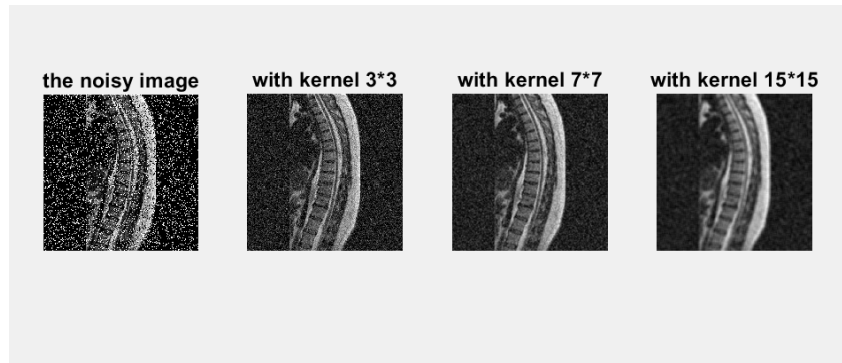


Figure 14: the effect of kernel size in grainy noised image

4.2 Gaussian Filter

Explore about Gaussian filter and write a function that applies a Gaussian filter to an input image. What is the effect of variance and kernel size on the output image?

GUSSIAN FILTER is a low pass filter which emit the high frequencies and intensities, in other words the noises and the image will become blurred. The Kernel coefficient in the Gaussian filter decreases the distance from the kernel centers. Generally, the size of the Gaussian filter is odd, and the kernel matrix is symmetrically applied.

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

- σ defines how "wide" or "narrow" the blur is. A higher sigma causes a stronger blur effect, while a lower sigma creates a milder blur.

$$\text{filteredimage}(i, j) = \sum_{x=-k}^k \sum_{y=-k}^k G(x, y) \cdot \text{image}(i + x, j + y)$$

The Gaussian function assigns the highest weight to the pixel at the center of the kernel, and the weight decreases as the distance from the center increases. This means that the center pixel contributes the most to the average, while the surrounding pixels have a progressively smaller impact. This property of the Gaussian kernel helps in smoothing the image while preserving important features such as edges.

As the result shows us the Gaussian filter performs better than the mean filter.

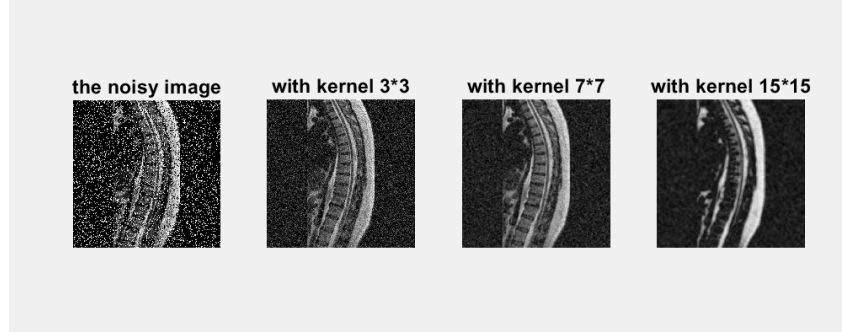


Figure 15: the effect of kernel size in noisy image

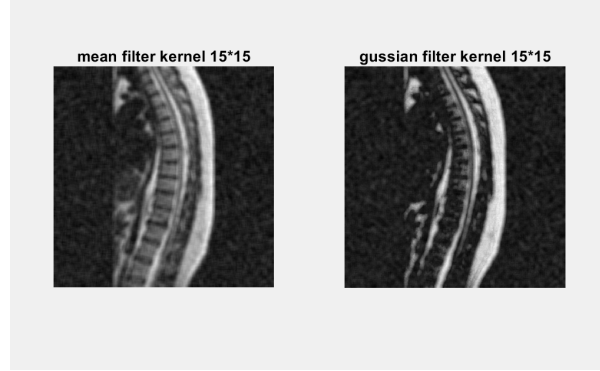


Figure 16: the difference between the Gaussian and Mean filter

4.3 Bilateral Filter

Explore about Bilateral filter and write a function that applies a Bilateral filter to an input image.

A bilateral filter is a nonlinear noise filtering and smoothing filter with one important difference from Gaussian filter which is the intensity of each pixel depends not only the Euclidean distance of pixels, but also on the radiometric differences. This difference is clear in its formula:

As the result shows us the Gaussian filter performs better than the mean filter.

$$I'(x, y) = \frac{1}{W(x, y)} \sum_{i=-k}^k \sum_{j=-k}^k I(x+i, y+j) \cdot G_s(i, j) \cdot G_r(I(x+i, y+j), I(x, y)) \quad (1)$$

where:

$$G_s(i, j) = \exp\left(-\frac{i^2 + j^2}{2\sigma_s^2}\right) \quad (2)$$

$$G_r(I, I') = \exp\left(-\frac{(I - I')^2}{2\sigma_r^2}\right) \quad (3)$$

$$W(x, y) = \sum_{i=-k}^k \sum_{j=-k}^k G_s(i, j) \cdot G_r(I(x+i, y+j), I(x, y)) \quad (4)$$

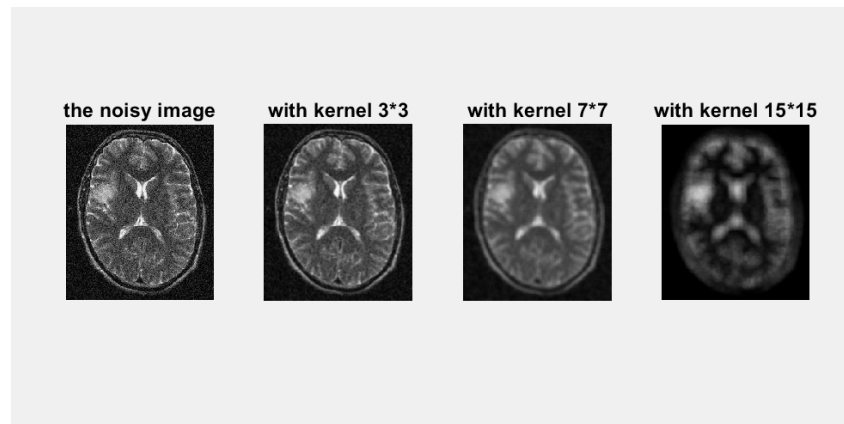


Figure 17: the effect of kernel size in noisy image

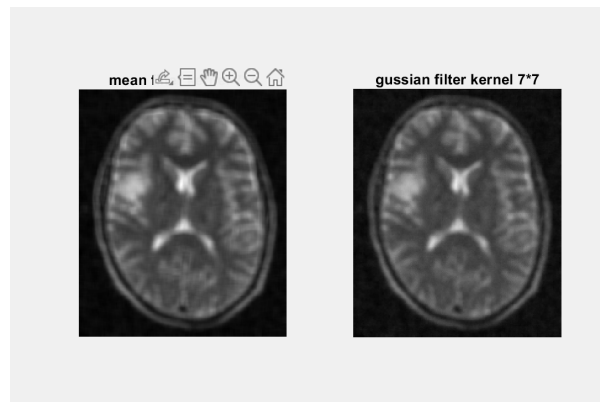


Figure 18: the difference between the Gaussian and Mean filter

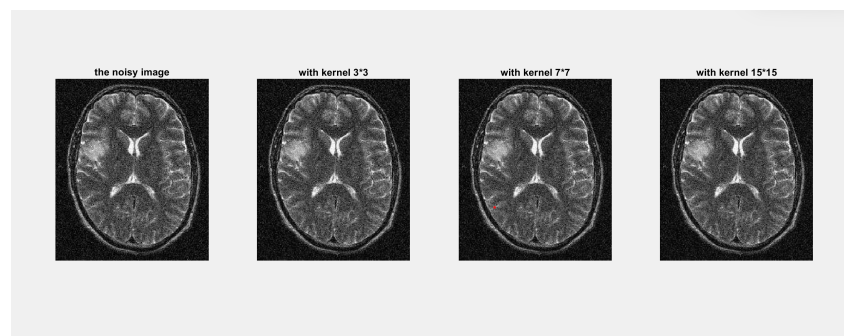


Figure 19: the effect of kernel size in noisy image

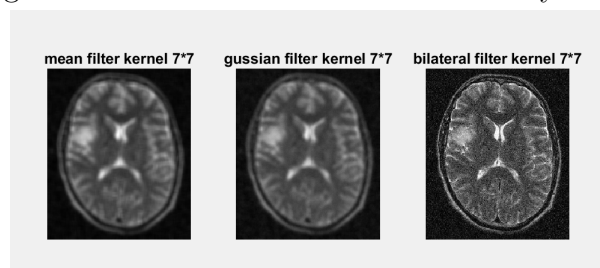


Figure 20: the difference between the Gaussian and Mean filter and bilateral

where:

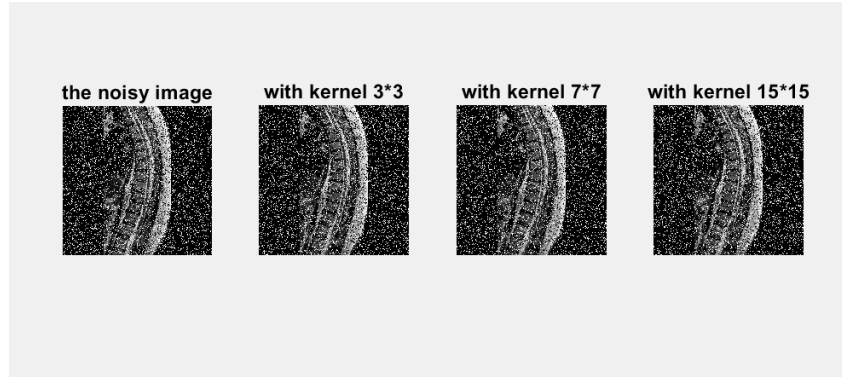


Figure 21: the effect of kernel size in noisy image

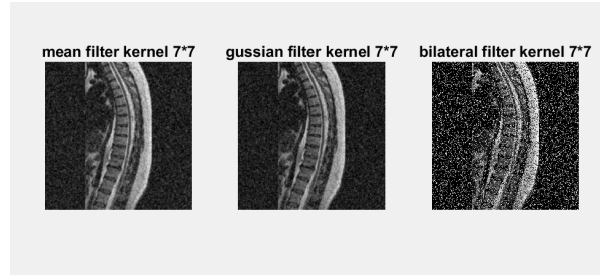


Figure 22: the difference between the Gaussian and Mean filter

- $I(x, y)$ is the original image intensity at pixel (x, y) .
- $I'(x, y)$ is the filtered intensity at pixel (x, y) .
- $G_s(i, j)$ is the spatial Gaussian kernel, controlled by σ_s .
- $G_r(I, I')$ is the range Gaussian kernel, controlled by σ_r .
- $W(x, y)$ is the normalization factor.

As the figures shows us the bilateral filters operate way better than Gaussian and mean filters in the tumor image but in the spine image because of having the salt and pepper noise the filter doesn't emit the noise because bilateral filter will keep the sharp changes in the pixels but in the other two filters they decrease the amount of brightness difference between neighbor pixels.

4.4 Evaluation

Evaluate the output of the three methods discussed in this section using SNR and PSNR criteria and present the results, compared with [original_brain_MRI.jpg](#) and [original_spine_MRI.jpg](#) in a table. Also, calculate these two criteria for the initial noisy image and declare which method performed better in noise reduction. Can you guess what kind of noise each picture has?

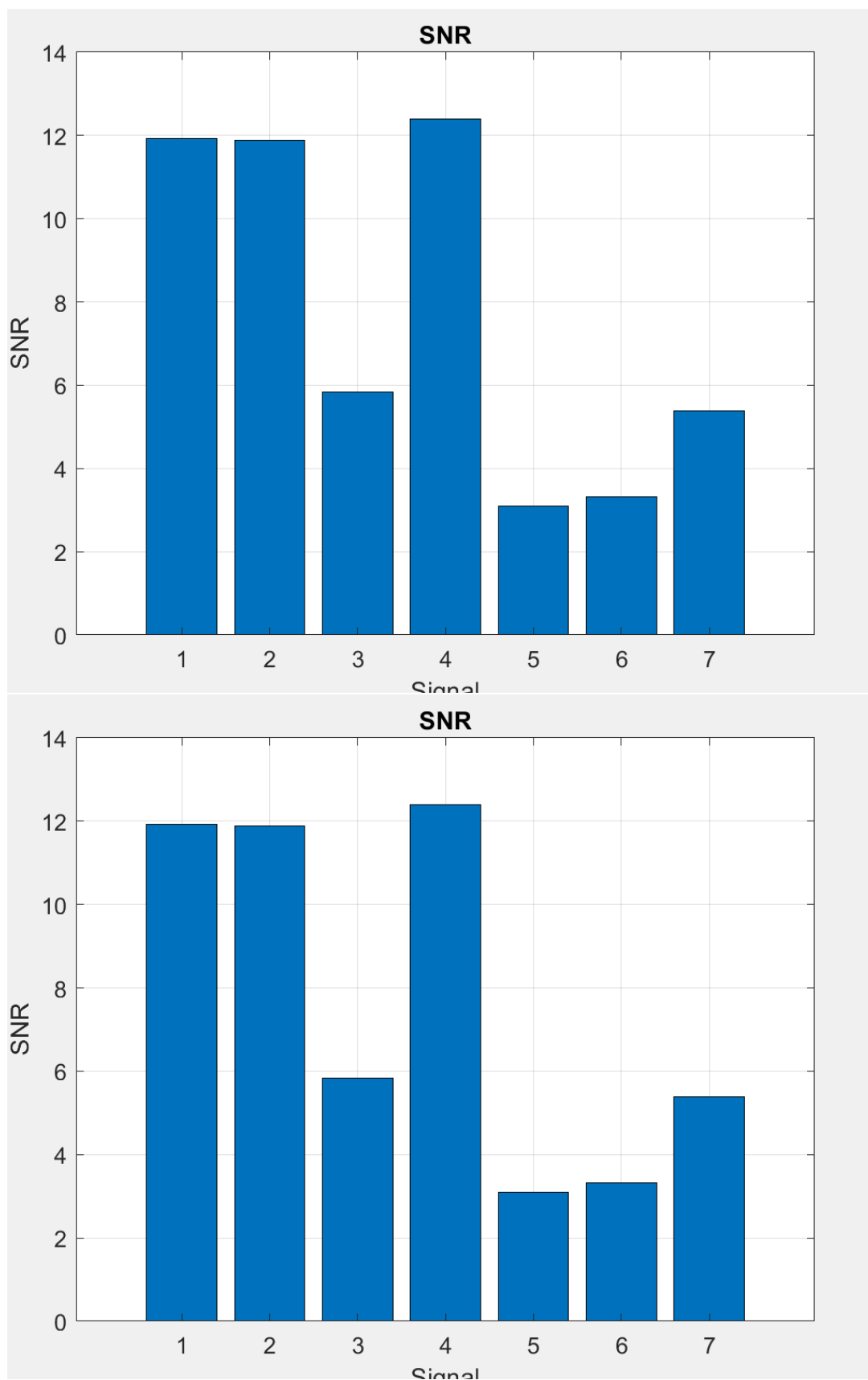


Figure 23: SNR and PSNR of the different filters

5 Adaptive Filter

5.1 Introduction

Adaptive filters are dynamic signal processing tools that adjust their parameters in real-time to accommodate varying input signal characteristics. Unlike fixed filters, which have static coefficients, adaptive filters modify their behavior based on the changing conditions of the data they process. This flexibility makes them particularly effective in environments where signal properties fluctuate, such as in biomedical image processing. In this field, adaptive filters are used for tasks like noise reduction, artifact removal, and image enhancement, ensuring that important features are preserved while unwanted disturbances are minimized. They employ algorithms, such as the Least Mean Squares (LMS) or Recursive Least Squares (RLS), to continuously optimize filter parameters based on feedback from the output. This ability to learn and adapt makes adaptive filters invaluable for improving image quality and accuracy in medical diagnostics and research applications.

5.2 Questions

1. What are finite impulse response (FIR) Adaptive Filters, and What are their specific applications in Biomedical Image Processing? Explore about Wiener filters.

FIR Adaptive Filter is a type of digital filter that continuously updates its coefficients based on incoming data to adapt to changing signal conditions. Unlike fixed filters, adaptive filters modify their behavior in real-time to optimize performance according to a predefined criterion, such as minimizing error.

$$y(n) = \sum_{k=0}^{M-1} w_k(n) \cdot x(n - K) \quad (5)$$

- $W_k(n)$ are the filter coefficients at time n ,
- M is the filter order (number of taps).

The adaptive algorithm modifies $W_k(n)$ to minimize an error function. The application of the adaptive filters in image processing:

- *MRI* :MRI images suffer from Gaussian noise due to thermal and electrical interference. Adaptive filtering helps suppress this noise while preserving image details.
- *Ultrasound – Speckle* : Ultrasound images suffer from speckle noise, which reduces image contrast and clarity. Adaptive FIR filters help reduce speckle noise using **Wiener filtering** or LMS-based adaptive filtering. Wiener filters adapt based on the local image variance, making them effective in preserving important details while reducing noise. The Wiener filter works in the **frequency domain** and is given by

$$H(u, v) = \frac{|F(u, v)|^2}{|F(u, v)|^2 + \frac{S_n(u, v)}{S_f(u, v)}} \quad (6)$$

where:

- $H(u, v)$ is the Wiener filter function.
- $|F(u, v)|^2$ is the power spectrum of the original signal.
- $S_n(u, v)$ is the noise power spectrum.
- $S_f(u, v)$ is the original image power spectrum.

In the spatial domain, the Wiener filter estimate $\hat{f}(x, y)$ is:

$$\hat{f}(x, y) = g(x, y) - \frac{\sigma_n^2}{\sigma_f^2 + \sigma_n^2}(g(x, y) - \mu) \quad (7)$$

where:

- $g(x, y)$ is the noisy image.
- μ and σ_f^2 are the local mean and variance.
- σ_n^2 is the noise variance.

The behavior of Wiener filtering depends on the local variance:

- If variance is low (smooth region) $\Rightarrow$ Stronger noise reduction.
- If variance is high (edge/detail) $\Rightarrow$ Weaker filtering to preserve details.

This adaptive approach makes Wiener filtering superior to simple smoothing filters.

2. Explore the role of Stochastic Gradient Descent(SGD) in optimizing filter coefficients in real-time, its convergence properties, and its application in adaptive filter design.

In adaptive filtering, the objective is to minimize the error between the desired signal $d(n)$ and the filter output $y(n)$. This is typically done by minimizing a cost function like the Mean Square Error (MSE):

$$J(w) = E [(d(n) - y(n))^2] \quad (8)$$

where:

- $d(n)$ is the desired output (clean signal).
- $y(n)$ is the adaptive filter output.
- w represents the filter coefficients.

SGD updates the filter coefficients using the gradient descent rule:

$$w(n+1) = w(n) - \eta \nabla J(w) \quad (9)$$

where:

- η is the learning rate (step size).

- $\nabla J(w)$ is the gradient of the cost function with respect to w .

Since computing the exact expectation $E[(d(n) - y(n))^2]$ is computationally expensive, SGD approximates the gradient using a single data sample:

$$w(n+1) = w(n) + \eta e(n)x(n) \quad (10)$$

where:

- $e(n) = d(n) - y(n)$ is the error signal.
- $x(n)$ is the input signal.
- η controls the step size and affects convergence.

Convergence Properties of SGD

SGD has certain properties that influence its performance in adaptive filtering:

- Large $\eta \rightarrow$ Faster convergence but risk of instability (oscillations).
- Small $\eta \rightarrow$ More stable but slow convergence.

3. Write a function that applies an Adaptive filter to an input image. Suggest a way to tune learning parameter. Apply this filter to [original_brain_MRI.jpg](#) and [original_spine_MRI.jpg](#).

4. Evaluate the output of this method using SNR and PSNR criteria and present the results. Which of these four filters did better for each image? Can you explain why?

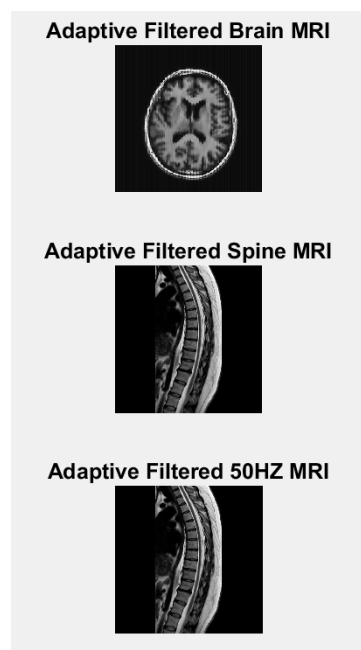


Figure 24: Gradient Descent Algorithm